READER NOTES

FUNDAMENTALS OF FINANCE | MODULE 9

Derivatives: Pricing, Strategies and Risk Management (Part 1)

STUDY NOTES · EDITION 1.0 · WEEK 9

How to use these notes

These notes cover the whole of Week 9: every concept and every worked example, written for reading and recall rather than for listening. They are yours to keep, mark up and revise from.

The week has nine topics in three parts. Part A defines what a derivative is and builds the call option contract from the ground up. Part B decomposes the premium and reads the payoff. Part C does the same for puts, which mirror calls at every point but one. Read the parts in order the first time; one running example — Zomato, with Rahul buying calls and Anjali buying puts — carries through all nine topics.

When assessment approaches you will not have time to re-read everything, and you will not need to. The Quick Recap boxes, the module summary and the self-check at the back were written for exactly that week. A workable plan: recaps first, then the summary, then attempt the self-check from memory before looking at the answers.

What each topic contains

Every topic follows the same rhythm, so you always know where to look.

Section	What it gives you
In one line	The whole idea compressed to a single sentence
Why this matters	Where the concept sits in the course, and why it earns your attention
The core idea	The full explanation, developed step by step
Figure	A diagram or comparison that makes the idea visible
Callouts	Key ideas, worked examples, pitfalls and exam pointers
Quick recap	Five bullets to revise from

Callouts used in this document

KEY IDEA

The most important takeaway of a section. If you remember nothing else, remember this.

COMMON PITFALL

A mistake learners make repeatedly, in assignments and in examinations.

EXAM POINTER

How the concept is typically tested, and what a full answer must contain.

REAL-WORLD EXAMPLE

A concrete case showing the concept at work outside the classroom.

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CONSOLIDATION

Module summary, by learning objective

Glossary and self-check

Module overview

This module introduces the first instrument in the course whose value is entirely borrowed. A derivative generates no cash flows of its own; its worth is a function of an underlying asset's price. The module builds the option contract field by field, works two trades through to their outcomes, decomposes the premium into its two components, and reads the payoff diagrams that professionals actually use.

One thread runs through all nine topics: **the premium buys a floor**. An option buyer is not purchasing the asset and not betting on a number — they are buying the right to walk away, and the premium is what that right costs.

Learning objectives

By the end of this module, you will be able to:

1. **Explain** what a derivative is and why an underlying asset is essential to its existence.
2. **Describe** the call option contract and work a trade through both possible outcomes.
3. **Decompose** an option premium into intrinsic and time value, and read a payoff diagram.
4. **Describe** put options, apply the protective and cash-secured put strategies, and compute put payoffs.

Topic map

9 topics, three parts. The Source column gives the lecture video each topic draws on, so you can go back to the original; the Serves column shows which objective it supports.

Topic	What it covers	Part	Source	Serves
1	What a derivative is	A · The contract	L32 V1	LO 1
2	The call option contract	A · The contract	L33 V1	LO 2
3	Trading a call: the two scenarios	A · The contract	L33 V2	LO 2
4	What an option premium is made of	B · Premium	L33 V3	LO 3
5	The long call payoff	B · Payoff	L33 V4	LO 3
6	The short call, and why anyone sells	B · Payoff	L33 V4	LO 3
7	The put option contract	C · Puts	L34 V1	LO 4
8	Protective put and cash-secured put	C · Puts	L34 V2	LO 4
9	Put premium and payoff	C · Puts	L34 V3–V4	LO 4

Key terms at a glance

Fuller definitions appear in the glossary at the back. Skim these before your first read; they carry most of the week's vocabulary.

Derivative
A contract that derives its value entirely from an underlying asset.

Underlying asset
The share, bond or index on which a derivative is written; without it the contract cannot exist.

Option
A derivative giving its buyer a right, but not an obligation.

Call option
The right to buy the underlying at the strike price.

Put option
The right to sell the underlying at the strike price.

Premium
The price of the option, paid upfront by the buyer to the seller.

Strike price
The predetermined price at which the option may be exercised; fixed at creation.

Spot price
The current market price of the underlying, changing continuously.

Expiry date
The date the buyer's right ends — conventionally the last Tuesday of the month.

Lot size
The fixed quantity of shares in one derivative contract.

Exercising
Using the right the contract grants; always a choice, never an obligation.

Long position
The buyer of an option, holding the right.

Short position
The seller, or writer, carrying the obligation.

Option chain
The exchange listing of all contracts on an underlying, across strikes and expiries.

Last traded price
The most recent market price of a particular option contract.

Intrinsic value
What exercising the option immediately would earn; floored at zero.

Time value
Compensation for the uncertainty remaining until expiry.

Near-month contract
The contract expiring in the current month; usually the most actively traded.

Payoff diagram
A plot of profit or loss against possible spot prices on expiry.

Kink
The point at which the payoff line changes direction — the strike price.

Break-even point
The expiry price at which the buyer neither gains nor loses.

Protective put
A put bought against shares already owned, acting as portfolio insurance.

Cash-secured put
A put sold by an investor willing to buy the shares at the strike price.

Zero-sum contract
One in which the buyer's gain exactly equals the seller's loss.

PART A · THE CONTRACT

A value borrowed from somewhere else

1 What a Derivative Is

PART A · THE CONTRACT · TOPIC 1 OF 9 · SERVES LEARNING OBJECTIVE 1

IN ONE LINE

A derivative has no value of its own. It borrows every rupee of its worth from an underlying asset.

Why this matters

Weeks 9 and 10 are built entirely on this one definition. Every instrument that follows — calls, puts, futures, straddles — is a variation on a contract whose value is borrowed rather than owned.

The house that starts everything

Priya wants a house priced at ₹85 lakh but is not ready to move in yet. She pays the broker ₹2 lakh for the guaranteed right to buy it at ₹85 lakh six months later. Six months on, either outcome works in her favour.

If the market rises to ₹1 crore, she exercises the agreement, buys at the pre-decided ₹85 lakh, and pockets the appreciation. If the market falls to ₹65 lakh, she walks away. The ₹2 lakh is lost, but she now buys the same house at roughly a quarter off.

What bought that flexibility? The ₹2 lakh. In derivatives language that is the **premium** — the price paid for the option to take or not take possession later.

The definition

Notice that the agreement has no value of its own; its worth depends entirely on the house. Had the house been worth ₹5 crore, the premium would have been far higher.

An option is a derivative instrument that derives its value from the price of another asset. Unlike the assets studied so far — which generate future cash flows that can be discounted — derivatives have no independent existence. They are contracts designed to perform specific functions when a future outcome is unfavourable, and for this course a "future outcome" means the future price of an asset.

Underlying assets include shares, bonds and stock market indices such as the Nifty 50 or the Sensex. The instruments studied are options and futures.

Every derivative needs an underlying

Car insurance cannot exist unless there is a car to insure. In the same way, the existence of an underlying asset is **essential** to the existence of a derivative contract. Because Zomato shares exist, a derivative contract based on Zomato shares can exist.

Two functions follow. **Protection against risk**: like insurance, an option shields the holder from an unfavourable movement, and like insurance it is not free. **Speculation**: an option lets an investor take a position on future price movements, including when they cannot afford the asset outright. If Zomato trades at ₹210 and you want 1,000 shares but lack ₹2,10,000, a derivative lets you take the position for far less.

THE INSURANCE ANALOGY, MADE EXACT

FIRE INSURANCE ON A WAREHOUSE	A CALL OPTION ON ZOMATO
UNDERLYING ASSET The furniture warehouse itself	UNDERLYING ASSET The Zomato share
PREMIUM The ₹50 lakh paid for five years of cover	PREMIUM The sum paid for the right to buy later
THE CONTRACT The fire insurance policy	THE CONTRACT The option contract
WOULD IT INSURE AN AIRCRAFT? No — the value and risk of the underlying set the price	WOULD IT COST THE SAME ON ANY SHARE? No — for the same reason

The nature and value of the underlying ***directly determine*** *the price of the contract. That is true of insurance and of every derivative in this course.*

Figure 1.1 Insurance and options are the same structure. Both price a contract off something else entirely.

KEY IDEA

Every valuation earlier in this course discounted an asset's own cash flows. A derivative generates none. Its entire worth is a *function of someone else's* price — which is why the word derivative is literal rather than figurative.

EXAM POINTER

Define a derivative as an instrument deriving its value from an underlying asset, and state that the underlying is essential to its existence. Name both functions — protection and speculation — with one example each. The premium should be identified as the price of the right, not as a fee.

QUICK RECAP · TOPIC 1

- A derivative has no independent value; it borrows all of it from an underlying asset.
- The premium is the price paid for the right to act later.
- Underlyings include shares, bonds and indices such as the Nifty 50.
- No underlying, no derivative — as with insurance and the thing insured.
- Two functions: protection against risk, and speculation on price movements.

2 The Call Option Contract

PART A · THE CONTRACT · TOPIC 2 OF 9 · SERVES LEARNING OBJECTIVE 1

IN ONE LINE

Seven fields define any option, and only one of them moves during the trading day.

Why this matters

The anatomy here is the vocabulary for everything that follows. Confusing strike price with premium, or lot size with quantity purchased, makes every later calculation impossible.

Rahul's position

At the beginning of the month Zomato trades at ₹260 per share. Rahul believes the quarterly results due at month end will push the price up sharply. He wants 1,000 shares, but that would require ₹2,60,000 and he does not have it. So he buys the right to purchase instead — a **call option**. Think of it as the right to *call* the shares towards you.

Lot size and expiry

In the cash market you can buy a single share. In the derivatives market you must trade the fixed quantity the exchange specifies, called a **lot**. Here the lot size is 1,000 shares.

A company can exist as long as it remains viable, but a derivative contract is created for a fixed period. Think of it as a cinema ticket for a particular Saturday: it gives you the right to watch the film that day but does not compel you to use it, and once Saturday passes the ticket is void.

Three monthly contracts are generally available at any time. The **near-month** expires on the last Tuesday of the current month and carries the highest trading activity; the **next-month** and **far-month** follow, and the cycle rolls forward continuously. If the last Tuesday is a market holiday, expiry moves to the preceding business day.

What can and cannot change

Only one number in the contract changes during the trading day — the **premium**. Strike price, expiry date and lot size are all fixed in advance by the exchange.

The ₹12,000 premium is Rahul's entire initial cash outflow, and also the **maximum he can lose**. That cap is what he bought.

SEVEN FIELDS · ONE OF THEM MOVES

Element	Value	What it means	Fixed or moving
Underlying asset	Zomato share	The contract cannot exist without it	Fixed
Spot price	₹260 per share	The current market price of the share	Moves continuously
Strike price	₹270 per share	The predetermined price at which Rahul may buy	Fixed at creation
Expiry date	Last Tuesday of the month	The date his right ends	Fixed at creation
Lot size	1,000 shares	The fixed quantity set by the exchange	Fixed at creation
Premium	₹12 per share — ₹12,000 total	The price paid for the right	**Moves continuously**
Position	Long call	Rahul is the buyer, holding the right	Fixed

*The premium is the buyer's **entire initial outflow and maximum possible loss**. Everything else about the contract was decided before he arrived.*

Figure 2.1 The call option contract. Only the spot price and the premium move.

COMMON PITFALL

Confusing the strike price with the price paid. Rahul pays ₹12 per share for the *right* to buy at ₹270. He does not pay ₹270 unless and until he exercises. The two numbers do very different jobs.

EXAM POINTER

Name all the contract elements and identify the premium as the only one that moves during the session. State the lot size convention and the near, next and far-month expiry cycle — last Tuesday of the month in the Indian market, moving to the preceding business day if that is a holiday.

QUICK RECAP · TOPIC 2

- A call option gives the right to buy the underlying at a fixed strike price.
- Derivatives trade in fixed lots set by the exchange — here, 1,000 shares.
- Contracts expire on the last Tuesday of the month, in near, next and far-month cycles.
- Strike, expiry and lot size are fixed at creation; only the premium moves.
- The premium is the buyer's entire outflow and maximum possible loss.

3 Trading a Call: The Two Scenarios

PART A · THE CONTRACT · TOPIC 3 OF 9 · SERVES LEARNING OBJECTIVE 2

IN ONE LINE

Whatever happens, Rahul's loss stops at ₹12,000 — and his gain does not stop at all.

Why this matters

The asymmetry demonstrated here is the entire commercial logic of buying options. It is also what makes the writer's position, in Topic 6, so different.

What exercising means

Exercising means using the right the contract grants — the same sense as exercising your right to vote. It is a choice, not an obligation, and Rahul will make it only if it is profitable.

Scenario one · results are strong

Zomato rises to ₹320. Rahul exercises and buys 1,000 shares at the strike price of ₹270 when they are worth ₹320. The gross gain is ₹50 per share; less the ₹12 premium, the net gain is **₹38 per share**, or **₹38,000** in total — on an initial commitment of ₹12,000.

Scenario two · results disappoint

Zomato falls to ₹230. It makes no sense to pay ₹270 for a share available at ₹230, so Rahul lets the option expire unexercised. The contract expires worthless and he takes no further action.

His loss is limited to the **₹12,000 premium** — not ₹30,000, not any larger figure. And if he still wants the shares he can simply buy them in the cash market at ₹230.

The central idea

Options protect the buyer from an unfavourable downward movement while allowing full benefit from a favourable upward one. If Zomato rose to ₹500 or ₹1,000, Rahul's profit would keep climbing. However far it falls, his loss stays at ₹12,000.

That is not a quirk of these particular numbers. It is the structural feature every option buyer is paying for.

TWO OUTCOMES, TWO VERY DIFFERENT SHAPES

SCENARIO 1 · PRICE RISES TO ₹320

ACTION
Exercise — buy at ₹270 what is worth ₹320

GROSS GAIN
₹50 per share

LESS PREMIUM
₹12 per share

NET GAIN
₹38 per share, or ₹38,000

IF IT ROSE FURTHER
The profit keeps climbing — no ceiling

SCENARIO 2 · PRICE FALLS TO ₹230

ACTION
Let it expire — no sense paying ₹270 for a ₹230 share

GROSS GAIN
Nil

LESS PREMIUM
₹12 per share

NET LOSS
₹12,000 — and no more

IF IT FELL FURTHER
The loss does not grow

*An initial commitment of **₹12,000** produced a ₹38,000 gain in one case and a ₹12,000 loss in the other. The upside is open-ended; the downside is capped.*

Figure 3.1 The two scenarios. Note that the loss is identical at ₹230, ₹200 or ₹0.

KEY IDEA

The premium buys a **floor**. That is the whole product. Rahul is not buying Zomato shares and not betting on a number — he is buying the right to walk away, and ₹12,000 is what that right costs.

EXAM POINTER

Work both scenarios with the arithmetic shown, and state that the loss in the second is capped at the premium regardless of how far the price falls. Note explicitly that the buyer may still purchase in the cash market afterwards — abandoning the option does not prevent acquiring the shares.

QUICK RECAP · TOPIC 3

- Exercising means using the right; it is a choice, never an obligation.
- At ₹320 Rahul exercises for a net gain of ₹38 per share, or ₹38,000.
- At ₹230 he lets the option lapse, losing only the ₹12,000 premium.
- The loss is identical however far the price falls; the gain has no ceiling.
- Options protect against downward movement while preserving the upside.

PART B · PREMIUM AND PAYOFF

What you pay, and what you get back

4 What an Option Premium Is Made Of

PART B · PREMIUM AND PAYOFF · TOPIC 4 OF 9 · SERVES LEARNING OBJECTIVE 3

IN ONE LINE

Every premium is intrinsic value plus time value — and viewing the trade from the seller's side is what makes both obvious.

Why this matters

This topic turns options from a narrative into a calculation. The decomposition here is the direct foundation of moneyness and every strategy in Week 10.

Where premiums come from

An **option chain** lists the contracts available on an underlying — several strike prices, all sharing the same expiry, each trading at its own market premium. Two columns matter: the strike price, and the last traded price, which changes continuously through the day just like a share price.

Three factors drive the premium: the spot price of the underlying, which moves continuously; the strike price, fixed when the contract was created; and market volatility, which reflects how far the share might move before expiry.

Intrinsic value, from the seller's side

View the transaction from the seller's position. Suppose the spot price is ₹255 and a buyer approaches with several proposals, each requiring the seller to deliver 1,000 shares in a month at a stated price.

At a strike of ₹200, the seller would immediately face a disadvantage of ₹55 per share — selling for ₹200 what is worth ₹255. That built-in disadvantage must be compensated, and it is called **intrinsic value**: the higher of (spot − strike) or zero for a call. It can never be negative.

Hence the inverse relationship: **low strike, high premium; high strike, low premium**. A lower strike gives the buyer the right to purchase at a more favourable price, so the seller demands more. Note that even where intrinsic value is zero the premiums still differ — a ₹260 strike needs the share to move only ₹5 to become exercisable, while a ₹320 strike needs a far larger move.

Time value

The second component is harder to calculate but simple to grasp. A call seller fears the price rising well above the strike. Put yourself in that position with a ₹270 strike and a ₹255 spot: would you be more concerned with 5 trading days remaining, 30 days, or 80?

Naturally the longer the time, the greater the chance the price moves against you. A longer period until expiry creates greater risk for the seller, and the greater the risk, the higher the premium demanded. **Time value** therefore reflects both the time remaining and the uncertainty about how the share may move in that period.

Option premium = intrinsic value + time value. Given any two, the third follows by subtraction.

TWO CONTRACTS, ONE SPOT PRICE OF ₹255

Call strike	Last traded price	Intrinsic value	Time value	Why
₹240	₹18.70	₹15.00	₹3.70	Spot exceeds strike by ₹15, so that much is built in
₹270	₹2.75	Nil	₹2.75	Strike above spot, so the entire premium is time value

*At a strike of ₹255 and above, intrinsic value is zero and the **whole premium is time value**. That is what a buyer is paying for: the chance that the price moves before expiry.*

Figure 4.1 Premium decomposition. Given any two of the three quantities, the third follows.

REAL-WORLD EXAMPLE

An option can be sold at any time before expiry, exactly like a share — so a trader need never touch the underlying. Rahul buys the ₹270 strike call at ₹2.75, paying ₹2,750 for the lot. Within a week Zomato rises to ₹300 and the premium reaches roughly ₹35. He sells the contract at ₹35 for a gain of ₹32.25 per share, or **₹32,250**. In the Indian market most option positions are closed this way rather than by exercising.

EXAM POINTER

Compute time value as premium minus intrinsic value — do not merely describe it. State the call intrinsic value formula with the zero floor, and give the directional rule: call premiums fall as strikes rise, and both call and put premiums rise with time to expiry. The reason — seller's risk — carries the explanation marks.

QUICK RECAP · TOPIC 4

- An option chain lists all strikes for one expiry, each with its own market premium.
- Three drivers: spot price, strike price and market volatility.
- Call intrinsic value = max(spot − strike, 0), and never negative.
- Low strike means high call premium; high strike means low premium.
- Premium = intrinsic value + time value; time value grows with time to expiry.

5 The Long Call Payoff

PART B · PREMIUM AND PAYOFF · TOPIC 5 OF 9 · SERVES LEARNING OBJECTIVE 3

IN ONE LINE

Flat below the strike, a kink at it, and a rising line above — with break-even sitting a full premium beyond.

Why this matters

The payoff diagram is how professionals think about every option position. Reading it correctly prevents the single most common error in options: assuming that crossing the strike means profit.

Drawn as though today were expiry

A payoff diagram is drawn as though today were the expiry date. Because the option has reached expiry its **time value is zero**, so the payoff depends only on intrinsic value and the premium paid. The horizontal axis shows possible spot prices on expiry; the vertical axis shows the resulting profit or loss.

Three features to read

The flat section. At or below the strike, the loss stays at ₹12 per share, or ₹12,000. However far the price falls, Rahul cannot lose more than the premium.

The kink. The line changes direction at the strike price of ₹270 and begins to rise. This is the bend that distinguishes an option from a futures contract, and it exists because the buyer can abandon the contract.

The break-even. The line crosses zero at ₹282. **Break-even for a long call = strike price + premium.**

The warning that goes with it

A share price above the strike does *not* by itself mean profit. At ₹275 the option has ₹5 of intrinsic value but cost ₹12, so Rahul is still ₹7 per share down. Profit begins only once the gain from exercising exceeds the premium paid.

This is the discriminator examiners use most often, because the intuition runs the wrong way: crossing the strike feels like success, and it is merely the point at which losses start shrinking.

THE LONG CALL PAYOFF · STRIKE ₹270, PREMIUM ₹12, LOT 1,000

Price on expiry	Intrinsic value	Net per share	Total	Zone
₹250	Nil	−₹12	−₹12,000	Flat — maximum loss
₹260	Nil	−₹12	−₹12,000	Flat — maximum loss
₹270 (strike)	Nil	−₹12	−₹12,000	The kink
₹275	₹5	−₹7	−₹7,000	Rising, still a loss
₹280	₹10	−₹2	−₹2,000	Rising, still a loss
₹282	₹12	Nil	Nil	**Break-even**
₹290	₹20	+₹8	+₹8,000	Profit
₹300	₹30	+₹18	+₹18,000	Profit
₹320	₹50	+₹38	+₹38,000	Profit

*At ₹275 the option is **in the money and still loss-making**. The premium must be recovered before profit begins — which is why break-even sits at ₹282, not ₹270.*

Figure 5.1 The long call payoff. Three rows sit above the strike and below break-even.

COMMON PITFALL

Treating "in the money" as equivalent to "profitable". They are different conditions. An option is in the money above the strike, but profitable only above strike plus premium. Between ₹270 and ₹282 it is both in the money and losing.

EXAM POINTER

Build the payoff table with intrinsic value, net per share and total. State the break-even formula and compute it. Describe all three features — flat section, kink, rising line — and be explicit that the flat section extends indefinitely leftwards at a constant loss.

QUICK RECAP · TOPIC 5

- A payoff diagram assumes expiry, so time value is zero and only intrinsic value remains.
- Below the strike the payoff is flat at the premium loss, however far the price falls.
- The line kinks at the strike of ₹270 and rises thereafter.
- Break-even for a long call = strike + premium = ₹282.
- Between ₹270 and ₹282 the option is in the money and still loss-making.

6 The Short Call, and Why Anyone Sells

PART B · PREMIUM AND PAYOFF · TOPIC 6 OF 9 · SERVES LEARNING OBJECTIVE 3

IN ONE LINE

Every rupee Rahul gains, someone else loses. The writer accepts unlimited risk for a capped reward — and the premium is why.

Why this matters

Understanding the writer's side completes the picture and explains where premiums come from. It also sets up the covered call strategy in Week 10.

The mirror image

Rahul takes a **long call** position. The counterparty who sells him the option takes a **short call** position, giving Rahul the right to buy 1,000 shares from her at ₹270 on expiry.

The short call payoff is the exact mirror image of the long call, obtained by flipping the diagram across the horizontal axis. The seller's maximum gain is the ₹12,000 premium, earned whenever the price stays at or below ₹270. Above ₹282 her losses grow without limit.

Why anyone accepts that

No trader would accept the seller's obligation without compensation. The **premium is the payment that persuades a trader to sell the option and accept the associated risk**. She is being paid, upfront and in cash, to carry an exposure that is bounded on one side and open on the other.

Sellers typically hold the opposite expectation — they believe the price will stay flat or fall, in which case the option expires worthless and they keep the entire premium.

Zero-sum

At every possible share price the buyer's profit equals the seller's loss and vice versa. The combined profit and loss of the two parties always adds to zero. **Options are zero-sum contracts.**

This is worth pausing on. Unlike a share, where both a buyer and a seller can profit over time as the company grows, an option transfers value between two parties. Nothing is created.

LONG AND SHORT · EXACT MIRROR IMAGES

	Long call · the buyer	Short call · the writer
Action	Purchases the option	Sells the option
Premium	Pays ₹12,000 upfront	Receives ₹12,000 upfront
Holds	The right, with no obligation	The obligation to honour the buyer's decision
Expectation	The share price will rise	The price will fall or stay below the strike
Maximum gain	Unlimited — rises with the price	Capped at the ₹12,000 premium
Maximum loss	Capped at the ₹12,000 premium	Unlimited — grows as the price rises
At ₹320 on expiry	+₹38,000	−₹38,000

At every price, one party's gain is the other's loss. ***Options are zero-sum contracts*** *— value is transferred, never created.*

Figure 6.1 The two sides of one contract. Read any row across and the signs reverse.

KEY IDEA

The premium is not arbitrary. It is the market-clearing price at which someone is willing to accept an unlimited, one-sided risk — which is why premiums rise with volatility and with time to expiry. Both make the writer's position more dangerous.

EXAM POINTER

Present the two positions as a comparison with at least five rows, and state the zero-sum property explicitly. The most-missed mark is explaining *why* anyone writes options — the premium compensates for accepting the obligation, and the writer expects the price to stay below the strike.

QUICK RECAP · TOPIC 6

- The buyer is long the call; the seller, or writer, is short it.
- The writer receives the premium and carries the obligation.
- Writer's maximum gain is the premium; maximum loss is unlimited.
- The premium is the compensation that persuades a trader to accept that obligation.
- At every price the buyer's gain equals the writer's loss — options are zero-sum.

PART C · PUT OPTIONS

The same logic, running the other way

7 The Put Option Contract

PART C · PUT OPTIONS · TOPIC 7 OF 9 · SERVES LEARNING OBJECTIVE 4

IN ONE LINE

The right to sell at a locked price. Arman's deal is Priya's in reverse — insurance in its purest form.

Why this matters

Puts mirror calls at every point, which makes them easy to learn and easy to confuse. The memory aid in this topic is worth more than it looks.

Arman's reverse deal

Same house, opposite worry. Arman owns a house worth ₹85 lakh and plans to sell in six months to fund his daughter's education abroad. Prices in his locality have started wobbling and he cannot afford to sell for less. He pays the broker ₹2 lakh, and the broker guarantees to buy the house in six months for ₹85 lakh, whatever the market says that day.

If prices crash to ₹65 lakh, Arman exercises and sells at the locked ₹85 lakh — the ₹2 lakh has saved him a ₹20 lakh wound. If prices climb to ₹1.05 crore, he walks away, lets the ₹2 lakh go, and sells in the open market at the higher price.

What those ₹2 lakh bought was not the right to buy — he already owns the house. It was the right to **sell** at a locked price. Priya's contract was a bet with a safety net; Arman's is insurance in its purest form.

The definition, and the memory aid

A put option gives its buyer the right, but not the obligation, to sell an underlying asset at a predetermined price on a predetermined date. Like a call it has no independent value and is bought by paying a premium.

The memory aid: a *call* gives you the right to **call** the shares towards you; a *put* gives you the right to **put** the shares onto the other party. You call things towards you and put things away.

Anjali's trade

Zomato trades at ₹260 and results are due. Unlike Rahul, Anjali expects them to disappoint. The traditional way to profit from a decline is to sell shares — but she does not own any. So she buys a put: strike ₹250, premium ₹10 per share, lot size 1,000, total outlay **₹10,000**.

If the price falls to ₹210 she exercises, selling at ₹250 what the market values at ₹210. The gross gain is ₹40 per share; net of the premium, **₹30 per share or ₹30,000**. She never needed to own the shares beforehand — on the exchange the difference is settled under the contract terms. If the price rises to ₹290 she lets it expire, losing only the ₹10,000.

CALL AND PUT · MIRROR IMAGES AT EVERY POINT

	Call option	Put option
The right it grants	To buy the underlying at the strike	To sell the underlying at the strike
Buyer expects	The price to rise	The price to fall
Intrinsic value	max(spot − strike, 0)	max(strike − spot, 0)
As the strike rises	Premium falls	Premium rises
Break-even	Strike + premium	Strike − premium
Buyer's maximum loss	The premium paid	The premium paid
Buyer's maximum gain	Unlimited — price can rise without bound	Large but bounded — price cannot fall below zero

*Every row mirrors, with one exception. The **maximum gain** differs, because a share price can rise without limit but cannot fall below zero.*

Figure 7.1 Calls and puts side by side. Learn the differences, not two separate instruments.

KEY IDEA

The one row that does *not* mirror is the most important. A call buyer's upside is unlimited because prices have no ceiling; a put buyer's is bounded because prices have a floor at zero. Every other property reverses cleanly.

EXAM POINTER

Give both intrinsic value formulas and both break-evens. State the asymmetry in maximum gain and explain it by the zero floor on prices. A question asking you to distinguish calls from puts wants the comparison drawn on at least five axes, not two definitions.

QUICK RECAP · TOPIC 7

- A put option gives the right, not the obligation, to sell at a predetermined strike price.
- Call: the right to call shares towards you. Put: the right to put shares onto the other party.
- Put intrinsic value = max(strike − spot, 0); break-even = strike − premium.
- At ₹210 Anjali nets ₹30 per share, or ₹30,000, on a ₹10,000 outlay.
- A put buyer's gain is large but bounded, because prices cannot fall below zero.

8 Protective Put and Cash-Secured Put

PART C · PUT OPTIONS · TOPIC 8 OF 9 · SERVES LEARNING OBJECTIVE 4

IN ONE LINE

The same contract serves two entirely different purposes depending on whether you buy it or sell it, and whether you already own the shares.

Why this matters

These are the first genuine strategies in the course — combinations rather than single positions — and they preview the financial engineering of Week 10.

Buying versus selling a put

Buying a put gives the right to sell at the strike. **Selling** a put creates an obligation: if the buyer exercises, the seller must purchase the underlying at the strike price — paying ₹250 per share even when the market price is ₹210.

The put seller therefore accepts the larger risk, and as always, an investor who accepts risk demands compensation. The premium is precisely that risk premium. One seller is simply a trader with the opposite expectation, who thinks the price will hold steady or rise; if it stays above the strike the option expires worthless and the seller keeps the whole ₹10,000.

The protective put · insurance for shares you own

An investor bought 1,000 Zomato shares long ago at ₹180 and they now trade at ₹260. She does not want to sell, believing in the long-term prospects, but fears the upcoming results. She buys a put at a ₹250 strike for ₹10 per share — ₹10,000 in total.

If the price falls to ₹210 her shares lose value, but the put lets her sell at ₹250, protecting her effective selling price. If the price rises to ₹290 the put expires worthless and the ₹10,000 is gone — but that is like paying car insurance in a year with no accident. The money bought certainty and protection for a month.

A put purchased against shares already owned is a protective put, and it acts as insurance for a share portfolio.

The cash-secured put · getting paid to wait

A long-term investor genuinely wants to own Zomato but thinks ₹260 is too high, and would happily pay ₹250. She *sells* a ₹250 strike put and receives ₹10 per share.

If the price stays above ₹250 the option expires worthless and she keeps ₹10,000 as income for waiting. If the price falls and the option is exercised, she buys 1,000 shares at ₹250 — the price she was willing to pay anyway — and because she also collected ₹10 per share, her **effective cost is ₹240**.

Either outcome is acceptable, which is what makes the strategy sensible rather than speculative.

ONE CONTRACT, TWO STRATEGIES

PROTECTIVE PUT · YOU BUY	CASH-SECURED PUT · YOU SELL
STARTING POSITION You already own 1,000 shares	**STARTING POSITION** You want the shares, but at ₹250 not ₹260
ACTION Buy a ₹250 put for ₹10 per share	**ACTION** Sell a ₹250 put and receive ₹10 per share
IF PRICE FALLS TO ₹210 Shares lose value, but you sell at ₹250	**IF PRICE STAYS ABOVE ₹250** Option lapses — you keep ₹10,000 as income
IF PRICE RISES TO ₹290 Put expires worthless — ₹10,000 spent on protection	**IF EXERCISED** You buy at ₹250; effective cost ₹240 after the premium
PURPOSE Insurance for a share portfolio	**PURPOSE** Getting paid to wait for your price

*The protective put **spends** a premium to buy certainty. The cash-secured put **collects** one in exchange for accepting an obligation you were happy to take anyway.*

Figure 8.1 Two strategies from one contract. The difference is whether you buy or sell it.

COMMON PITFALL

Calling a protective put a wasted expense when the price rises and the option expires worthless. That is insurance working as designed. The premium bought a month of certainty; a year with no accident does not make motor insurance a mistake.

EXAM POINTER

Distinguish the two by *who holds what*: the protective put buyer already owns shares and pays for downside cover; the cash-secured put seller wants shares and is paid for agreeing to buy them lower. Compute the effective cost in the second case — strike minus premium received.

QUICK RECAP · TOPIC 8

- Selling a put creates an obligation to buy at the strike if exercised.
- A protective put is bought against shares already owned — insurance for a portfolio.
- If the price rises, the protective put expires worthless; the premium bought certainty.
- A cash-secured put is sold by an investor happy to buy at the strike.
- If exercised, the effective cost is strike minus premium: ₹250 − ₹10 = ₹240.

9 Put Premium and Payoff

PART C · PUT OPTIONS · TOPIC 9 OF 9 · SERVES LEARNING OBJECTIVE 4

IN ONE LINE

Put premiums rise as strikes rise — the exact opposite of calls — and the payoff has a floor where the call has none.

Why this matters

The closing topic of the week completes the mirror. Getting the directional rule right here is what prevents the most common error in reading an option chain.

The premium, from the seller's side again

The logic mirrors the call, with the seller's fear reversed. Approach a put seller with the spot at ₹255 and several proposals, each obliging them to *buy* 1,000 shares from you in a month at a stated price.

At a strike of ₹200 they are comfortable — buying at ₹200 what is worth ₹255. At ₹290 they would be agreeing to pay ₹35 above market; at ₹320, ₹65 above. They would agree only if compensated for the disadvantage already built into the contract.

Put intrinsic value = max(strike − spot, 0) — the exact opposite of a call. A higher strike gives the put buyer a greater built-in advantage, so **put premiums increase as strike prices increase**, while call premiums fall.

Time value works identically with the direction reversed: a call seller fears the price rising far above the strike, a put seller fears it falling far below. Longer time to expiry means more chance of an adverse move, so a higher premium.

Trading the put before expiry

Anjali buys the ₹240 strike put at ₹2.30 per share, paying ₹2,300 for the lot. Within a week negative news drops Zomato from ₹255 to ₹220. Intrinsic value alone is now ₹20 per share, and with time still remaining the premium rises to roughly ₹24. She sells the contract for a gain of ₹21.70 per share, or **₹21,700** — without ever buying or selling a Zomato share.

The payoff, and the one thing that differs

Read the put payoff from right to left. At or above the strike, Anjali does not exercise and her loss stays at the ₹10 premium, however high the price climbs. The line kinks at ₹250 and rises as the price falls below it, crossing zero at ₹240. **Break-even for a long put = strike − premium.**

One important difference from the call: a share price can rise without limit, but it cannot fall below zero. So the put buyer's maximum gain is large but *bounded*. In the extreme case the share falls to zero and Anjali sells worthless shares for ₹250 — a gain of ₹240 per share after the premium, or **₹2,40,000**. That is also the put writer's maximum loss.

THE LONG PUT PAYOFF · STRIKE ₹250, PREMIUM ₹10, LOT 1,000

Price on expiry	Intrinsic value	Net per share	Total	Zone
₹0 (worst case)	₹250	+₹240	+₹2,40,000	Maximum possible gain
₹150	₹100	+₹90	+₹90,000	Profit
₹210	₹40	+₹30	+₹30,000	Profit
₹230	₹20	+₹10	+₹10,000	Profit
₹240	₹10	Nil	Nil	**Break-even**
₹245	₹5	−₹5	−₹5,000	In the money, still a loss
₹250 (strike)	Nil	−₹10	−₹10,000	The kink
₹290	Nil	−₹10	−₹10,000	Flat — maximum loss

*As with the call, a price beyond the strike is **not yet a profit**. At ₹245 the option is worth ₹5 but cost ₹10. And the top row is the ceiling — prices cannot go below zero.*

Figure 9.1 The long put payoff. Read it right to left; the profitable region is on the left.

KEY IDEA

Call and put premiums move in **opposite directions** across an option chain. Reading down a column of strikes, call prices fall and put prices rise. If both move the same way, you are reading the chain wrongly.

EXAM POINTER

Build the payoff table and identify break-even as strike minus premium. State the bounded maximum gain and compute it: (strike − premium) × lot size. Give the directional rule for premiums across strikes — it is the single most reliable way to show you have understood the chain rather than memorised one example.

QUICK RECAP · TOPIC 9

- Put intrinsic value = max(strike − spot, 0), the opposite of a call.
- Put premiums rise as strikes rise, while call premiums fall.
- Time value grows with time to expiry for both, because the seller's risk grows.
- Break-even for a long put = strike − premium = ₹240.
- Maximum gain is bounded at (strike − premium) × lot = ₹2,40,000, because prices cannot fall below zero.

Module summary, by learning objective

Organised the way assessment will test you, rather than in topic sequence.

LO	What you should now be able to do	Where it was covered
1	Explain what a derivative is and why the underlying is essential	Topic 1 · Figure 1.1
2	Describe the call contract and work a trade through both outcomes	Topics 2 and 3 · Figures 2.1, 3.1
3	Decompose a premium and read call payoff diagrams	Topics 4, 5 and 6 · Figures 4.1, 5.1, 6.1
4	Describe puts, apply put strategies and compute put payoffs	Topics 7, 8 and 9 · Figures 7.1, 8.1, 9.1

The module in six sentences

A derivative has no value of its own — it borrows every rupee of its worth from an underlying asset, which is why no underlying means no contract. An option grants a right rather than an obligation, defined by seven fields of which only the premium moves during the trading day. Rahul's call on Zomato earns ₹38,000 if the share reaches ₹320 and loses exactly ₹12,000 however far it falls, which is the asymmetry the premium buys. Every premium decomposes into intrinsic value, the advantage already built into the contract, and time value, the seller's compensation for the uncertainty remaining. Payoff diagrams show a flat section, a kink at the strike and a rising line beyond — with break-even a full premium past the strike, so that a price above ₹270 is not yet a profit until it passes ₹282. Puts mirror calls at every point except one: a put buyer's gain is bounded, because a share price cannot fall below zero.

HOW THE NINE TOPICS CONNECT

PART A · THE CONTRACT

1 · What a derivative is
2 · The call contract
3 · Two scenarios

TAKEAWAY
A value borrowed, not owned

PART B · PREMIUM AND PAYOFF

4 · The premium
5 · Long call payoff
6 · The short call

TAKEAWAY
What you pay buys a floor

PART C · PUT OPTIONS

7 · The put contract
8 · Two strategies
9 · Premium and payoff

TAKEAWAY
The mirror, with one exception

The premium buys a floor. *An option buyer is not purchasing the asset and not betting on a number — they are buying the right to walk away.*

Figure M1 How the nine topics connect. Revise along the parts, not down the list.

Glossary and self-check

Term	Definition
Derivative	A contract that derives its value entirely from an underlying asset.
Underlying asset	The share, bond or index on which a derivative is written; without it the contract cannot exist.
Option	A derivative giving its buyer a right, but not an obligation.
Call option	The right to buy the underlying at the strike price.
Put option	The right to sell the underlying at the strike price.
Premium	The price of the option, paid upfront by the buyer to the seller.
Strike price	The predetermined price at which the option may be exercised; fixed at creation.
Spot price	The current market price of the underlying, changing continuously.
Expiry date	The date the buyer's right ends — conventionally the last Tuesday of the month.
Lot size	The fixed quantity of shares in one derivative contract.
Exercising	Using the right the contract grants; always a choice, never an obligation.
Long position	The buyer of an option, holding the right.
Short position	The seller, or writer, carrying the obligation.
Option chain	The exchange listing of all contracts on an underlying, across strikes and expiries.
Last traded price	The most recent market price of a particular option contract.
Intrinsic value	What exercising the option immediately would earn; floored at zero.
Time value	Compensation for the uncertainty remaining until expiry.
Near-month contract	The contract expiring in the current month; usually the most actively traded.
Payoff diagram	A plot of profit or loss against possible spot prices on expiry.
Kink	The point at which the payoff line changes direction — the strike price.
Break-even point	The expiry price at which the buyer neither gains nor loses.
Protective put	A put bought against shares already owned, acting as portfolio insurance.
Cash-secured put	A put sold by an investor willing to buy the shares at the strike price.

Term	Definition
Zero-sum contract	One in which the buyer's gain exactly equals the seller's loss.

Self-check

Ungraded retrieval practice. Answer from memory first, then check.

Q1 A call has a strike of ₹270 and a premium of ₹12. The share closes at ₹275 on expiry. Has the buyer profited?

> **No — there is a loss of ₹7 per share.** Intrinsic value is ₹5, but ₹12 was paid. Break-even is strike plus premium, or ₹282. Between ₹270 and ₹282 the option is in the money and still loss-making — in the money and profitable are different conditions.

Q2 The spot price is ₹255. A ₹270 put trades at ₹18.40. What is its time value, and what would the ₹270 call's intrinsic value be?

> **Time value ₹3.40; the call has no intrinsic value.** Put intrinsic = max(270 − 255, 0) = ₹15, so time value = 18.40 − 15.00. The call at the same strike is out of the money, since spot is below strike — the mirror behaviour.

Q3 Why is a call buyer's maximum gain unlimited while a put buyer's is not?

> **Because a share price can rise without limit but cannot fall below zero.** The most a put can be worth is the strike itself, less the premium — here (250 − 10) × 1,000 = ₹2,40,000. A call has no equivalent ceiling.

Q4 An investor owns 1,000 shares at ₹260 and buys a ₹250 put for ₹10. The price rises to ₹290 and the put expires worthless. Was the ₹10,000 wasted?

> **No — that is insurance working as designed.** The premium bought a month of downside certainty. A year with no accident does not make motor insurance a mistake. This is a protective put, and its purpose is protection, not profit.

Q5 An investor sells a ₹250 put for ₹10 and is later exercised. What did the shares actually cost?

> **₹240 per share.** She must buy at the ₹250 strike, but she collected ₹10 in premium, so her effective cost is ₹240. This is a cash-secured put, and it makes sense only for an investor who genuinely wanted the shares at that price.

Q6 Reading down an option chain from low strikes to high, what happens to call and put premiums?

> **Call premiums fall; put premiums rise.** A lower strike favours a call buyer, who gains the right to buy cheaply, so the seller demands more. A higher strike favours a put buyer, who gains the right to sell dearly. If both appear to move the same way, the chain is being read wrongly.